DevOps Lifecycle Architecture Diagram

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1. End-to-End DevOps Delivery Flow

The project operationalizes an automated 8-stage continuous delivery lifecycle:

1. **Plan:** Agile product backlog tracking 15 user stories and tasks.
2. **Code:** Java Spring Boot microservice versioned in Git using GitFlow.
3. **Build:** Apache Maven cleans, compiles source bytecode, and packages JAR.
4. **Test:** Selenium and JUnit execute 6 Critical User Journeys acting as quality gate.
5. **Package:** Standalone executable JAR archived in Jenkins artifact store.
6. **Release:** Docker container image built, tagged with build number and latest.
7. **Deploy:** Ansible playbook provisions target node and launches container.
8. **Operate & Monitor:** Automated HTTP /api/health probes monitor liveness.

2. Architectural Quality Assurance

If any test fails during Stage 4, the Jenkins quality gate halts immediately, preventing defective software from ever progressing to packaging or production deployment.